

OSSP SKILL 7

Terminal 1:

```
root@LAPTOP-0PA61QEN:~# ls /home
alurusaranya
root@LAPTOP-0PA61QEN:~# su alurusaranya
alurusaranya@LAPTOP-0PA61QEN:/root$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ mkdir -p ~/skill7
alurusaranya@LAPTOP-0PA61QEN:~$ cd ~/skill7
alurusaranya@LAPTOP-0PA61QEN:~/skill7$ nano memory.c
alurusaranya@LAPTOP-0PA61QEN:~/skill7$ gcc memory.c -o memory
alurusaranya@LAPTOP-0PA61QEN:~/skill7$ ./memory
Process ID: 43012
Code address : 0x5882285c7209
Global address : 0x5882285ca010
Static address : 0x5882285ca014
Static local : 0x5882285ca018
Heap address : 0x58825cd3d2a0
Stack address : 0x7ffc256bc32c
```

Press Enter to exit...

```
alurusaranya@LAPTOP-0PA61QEN:~/skill7$ ./memory
Process ID: 43048
Code address : 0x5a6f08b53209
Global address : 0x5a6f08b56010
Static address : 0x5a6f08b56014
Static local : 0x5a6f08b56018
Heap address : 0x5a6f0e7512a0
Stack address : 0x7ffff4b35e3c
```

Press Enter to exit...

```
GNU nano 7.2 memory.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void code_function()
{
}

int main()
{
    int stack_var = 30;
    static int static_local = 40;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 50;

    printf("Process ID: %d\n", getpid());

    printf("Code address : %p\n", (void *)code_function);
    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Static local : %p\n", (void *)&static_local);
    printf("Heap address : %p\n", (void *)heap_var);
    printf("Stack address : %p\n", (void *)&stack_var);

    printf("\nPress Enter to exit...\n");
    getchar();

    free(heap_var);

    return 0;
}
```

Terminal 2:

```
root@LAPTOP-0PA61QEN:~# su alurusaranya
alurusaranya@LAPTOP-0PA61QEN:/root$ cd ~/skill7
alurusaranya@LAPTOP-0PA61QEN:~/skill7$ cat /proc/43048/maps
5a6f08b52000-5a6f08b53000 r--p 00000000 08:30 10515 /home/alurusaranya/skill7/memory
5a6f08b53000-5a6f08b54000 r-xp 00001000 08:30 10515 /home/alurusaranya/skill7/memory
5a6f08b54000-5a6f08b55000 r--p 00002000 08:30 10515 /home/alurusaranya/skill7/memory
5a6f08b55000-5a6f08b56000 r--p 00002000 08:30 10515 /home/alurusaranya/skill7/memory
5a6f08b56000-5a6f08b57000 rw-p 00003000 08:30 10515 /home/alurusaranya/skill7/memory
5a6f0e751000-5a6f0e772000 rw-p 00000000 00:00 0 [heap]
7733ae600000-7733ae628000 r--p 00000000 08:30 51149 /usr/lib/x86_64-linux-gnu/libc.so.6
7733ae628000-7733ae7b1000 r-xp 00028000 08:30 51149 /usr/lib/x86_64-linux-gnu/libc.so.6
7733ae7b1000-7733ae800000 r--p 001b1000 08:30 51149 /usr/lib/x86_64-linux-gnu/libc.so.6
7733ae800000-7733ae804000 r--p 001ff000 08:30 51149 /usr/lib/x86_64-linux-gnu/libc.so.6
7733ae804000-7733ae806000 rw-p 00203000 08:30 51149 /usr/lib/x86_64-linux-gnu/libc.so.6
7733ae806000-7733ae813000 rw-p 00000000 00:00 0
7733ae8f2000-7733ae8f5000 rw-p 00000000 00:00 0
7733ae8fa000-7733ae8fc000 rw-p 00000000 00:00 0
7733ae8fc000-7733ae900000 r--p 00000000 00:00 0 [vvar]
7733ae900000-7733ae902000 r--p 00000000 00:00 0 [vvar_vclock]
7733ae902000-7733ae904000 r-xp 00000000 00:00 0 [vdso]
7733ae904000-7733ae905000 r--p 00000000 08:30 51146 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7733ae905000-7733ae930000 r-xp 00001000 08:30 51146 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7733ae930000-7733ae93a000 r--p 0002c000 08:30 51146 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7733ae93a000-7733ae93c000 r--p 00036000 08:30 51146 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7733ae93c000-7733ae93e000 rw-p 00038000 08:30 51146 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffff4b17000-7ffff4b38000 rw-p 00000000 00:00 0 [stack]
```

```
alurusaranya@LAPTOP-0PA61QEN:~/skill7$ pmap 43048
```

```
43048: ./memory
00005a6f08b52000      4K r---- memory
00005a6f08b53000      4K r-x-- memory
00005a6f08b54000      4K r---- memory
00005a6f08b55000      4K r---- memory
00005a6f08b56000      4K rw--- memory
00005a6f0e751000    132K rw--- [ anon ]
00007733ae600000    160K r---- libc.so.6
00007733ae620000   1572K r-x-- libc.so.6
00007733ae7b1000    316K r---- libc.so.6
00007733ae800000     16K r---- libc.so.6
00007733ae804000      8K rw--- libc.so.6
00007733ae806000     52K rw--- [ anon ]
00007733ae8f2000     12K rw--- [ anon ]
00007733ae8fa000      8K rw--- [ anon ]
00007733ae8fc000     16K r---- [ anon ]
00007733ae900000      8K r---- [ anon ]
00007733ae902000      8K r-x-- [ anon ]
00007733ae904000      4K r---- ld-linux-x86-64.so.2
00007733ae905000    172K r-x-- ld-linux-x86-64.so.2
00007733ae930000     40K r---- ld-linux-x86-64.so.2
00007733ae93a000      8K r---- ld-linux-x86-64.so.2
00007733ae93c000      8K rw--- ld-linux-x86-64.so.2
00007ffff4b17000    132K rw--- [ stack ]
total                2692K
```

```
total                2692K
```

```
alurusaranya@LAPTOP-0PA61QEN:~/skill7$ cat /proc/43048/smaps
```

```
5a6f08b52000-5a6f08b53000 r--p 00000000 08:30 10515
```

```
/home/alurusaranya/skill7/memory
```

```
Size:                4 kB
KernelPageSize:      4 kB
MMUPageSize:         4 kB
Rss:                 4 kB
Pss:                 4 kB
Pss_Dirty:           0 kB
Shared_Clean:        0 kB
Shared_Dirty:        0 kB
Private_Clean:       4 kB
Private_Dirty:       0 kB
Referenced:          4 kB
Anonymous:           0 kB
KSM:                 0 kB
LazyFree:            0 kB
AnonHugePages:       0 kB
ShmemPmdMapped:      0 kB
FilePmdMapped:       0 kB
Shared_Hugetlb:      0 kB
Private_Hugetlb:     0 kB
Swap:                0 kB
SwapPss:             0 kB
Locked:              0 kB
THPeligible:         0
```

```
VmFlags: rd mr mw me sd
```

```
5a6f08b53000-5a6f08b54000 r-xp 00001000 08:30 10515
```

```
/home/alurusaranya/skill7/memory
```

```
Size:                4 kB
KernelPageSize:      4 kB
MMUPageSize:         4 kB
Rss:                 4 kB
Pss:                 4 kB
Pss_Dirty:           0 kB
Shared_Clean:        0 kB
```

```

ShmemPmdMapped:    0 kB
FilePmdMapped:    0 kB
Shared_Hugetlb:    0 kB
Private_Hugetlb:   0 kB
Swap:              0 kB
SwapPss:           0 kB
Locked:            0 kB
THPEligible:       0
VmFlags: rd ex mr mw me sd
5a6f08b54000-5a6f08b55000 r--p 00002000 08:30 10515      /home/alurusaranya/skill7/memory
Size:              4 kB
KernelPageSize:    4 kB
MMUPageSize:       4 kB
Rss:               4 kB
Pss:               4 kB
Pss_Dirty:         0 kB
Shared_Clean:      0 kB
Shared_Dirty:      0 kB
Private_Clean:     4 kB
Private_Dirty:     0 kB
Referenced:        4 kB
Anonymous:         0 kB
KSM:               0 kB
LazyFree:          0 kB
AnonHugePages:     0 kB
ShmemPmdMapped:    0 kB
FilePmdMapped:    0 kB
Shared_Hugetlb:    0 kB
Private_Hugetlb:   0 kB
Swap:              0 kB
SwapPss:           0 kB
Locked:            0 kB
THPEligible:       0
VmFlags: rd mr mw me sd
5a6f08b55000-5a6f08b56000 r--p 00002000 08:30 10515      /home/alurusaranya/skill7/memory
Size:              4 kB
KernelPageSize:    4 kB

```

```

MMUPageSize:       4 kB
Rss:               0 kB
Pss:               0 kB
Pss_Dirty:         0 kB
Shared_Clean:      0 kB
Shared_Dirty:      0 kB
Private_Clean:     0 kB
Private_Dirty:     0 kB
Referenced:        0 kB
Anonymous:         0 kB
KSM:               0 kB
LazyFree:          0 kB
AnonHugePages:     0 kB
ShmemPmdMapped:    0 kB
FilePmdMapped:    0 kB
Shared_Hugetlb:    0 kB
Private_Hugetlb:   0 kB
Swap:              0 kB
SwapPss:           0 kB
Locked:            0 kB
THPEligible:       0
VmFlags: rd mr pf io de dd sd
7733ae902000-7733ae904000 r-xp 00000000 00:00 0          [vdso]
Size:              8 kB
KernelPageSize:    4 kB
MMUPageSize:       4 kB
Rss:               4 kB
Pss:               0 kB
Pss_Dirty:         0 kB
Shared_Clean:      4 kB
Shared_Dirty:      0 kB
Private_Clean:     0 kB
Private_Dirty:     0 kB
Referenced:        4 kB
Anonymous:         0 kB
KSM:               0 kB
LazyFree:          0 kB

```

```
Shared_Hugetlb:      0 kB
Private_Hugetlb:    0 kB
Swap:               0 kB
SwapPss:            0 kB
Locked:             0 kB
THPeligible:         0
VmFlags: rd ex mr mw me de sd
7733ae904000-7733ae905000 r--p 00000000 08:30 51146      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
Size:                4 kB
KernelPageSize:     4 kB
MMUPageSize:        4 kB
Rss:                4 kB
Pss:                0 kB
Pss_Dirty:          0 kB
Shared_Clean:       4 kB
Shared_Dirty:       0 kB
Private_Clean:      0 kB
Private_Dirty:      0 kB
Referenced:         4 kB
Anonymous:          0 kB
KSM:                0 kB
LazyFree:           0 kB
AnonHugePages:      0 kB
ShmemPmdMapped:     0 kB
FilePmdMapped:      0 kB
Shared_Hugetlb:     0 kB
Private_Hugetlb:    0 kB
Swap:               0 kB
SwapPss:            0 kB
Locked:             0 kB
THPeligible:         0
VmFlags: rd mr mw me sd
7733ae905000-7733ae930000 r-xp 00001000 08:30 51146      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
Size:                172 kB
KernelPageSize:     4 kB
MMUPageSize:        4 kB
Rss:                172 kB
```

```
KSM:                0 kB
LazyFree:           0 kB
AnonHugePages:      0 kB
ShmemPmdMapped:     0 kB
FilePmdMapped:      0 kB
Shared_Hugetlb:     0 kB
Private_Hugetlb:    0 kB
Swap:               0 kB
SwapPss:            0 kB
Locked:             0 kB
THPeligible:         0
VmFlags: rd mr mw me sd
7733ae905000-7733ae930000 r-xp 00001000 08:30 51146      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
Size:                172 kB
KernelPageSize:     4 kB
MMUPageSize:        4 kB
Rss:                172 kB
Pss:                24 kB
Pss_Dirty:          0 kB
Shared_Clean:       172 kB
Shared_Dirty:       0 kB
Private_Clean:      0 kB
Private_Dirty:      0 kB
Referenced:         172 kB
Anonymous:          0 kB
KSM:                0 kB
LazyFree:           0 kB
AnonHugePages:      0 kB
ShmemPmdMapped:     0 kB
FilePmdMapped:      0 kB
Shared_Hugetlb:     0 kB
Private_Hugetlb:    0 kB
Swap:               0 kB
SwapPss:            0 kB
Locked:             0 kB
THPeligible:         0
VmFlags: rd ex mr mw me sd
```

```
Shared_Clean:      0 kB
Shared_Dirty:      0 kB
Private_Clean:     0 kB
Private_Dirty:     8 kB
Referenced:        8 kB
Anonymous:         8 kB
KSM:               0 kB
LazyFree:          0 kB
AnonHugePages:     0 kB
ShmemPmdMapped:    0 kB
FilePmdMapped:     0 kB
Shared_Hugetlb:    0 kB
Private_Hugetlb:   0 kB
Swap:              0 kB
SwapPss:           0 kB
Locked:            0 kB
THPeligible:       0
VmFlags: rd wr mr mw me ac sd
7ffff4b17000-7ffff4b38000 rw-p 00000000 00:00 0      [stack]
Size:              132 kB
KernelPageSize:    4 kB
MMUPageSize:       4 kB
Rss:               16 kB
Pss:               16 kB
Pss_Dirty:         16 kB
Shared_Clean:      0 kB
Shared_Dirty:      0 kB
Private_Clean:     0 kB
Private_Dirty:     16 kB
Referenced:        16 kB
Anonymous:         16 kB
KSM:               0 kB
LazyFree:          0 kB
AnonHugePages:     0 kB
ShmemPmdMapped:    0 kB
FilePmdMapped:     0 kB
Shared_Hugetlb:    0 kB
```

```
alurusaranya@LAPTOP-0PA61QEN:~/skill7$ cat /proc/43048/status
```

```
Name: memory
Umask: 0002
State: S (sleeping)
Tgid: 43048
Ngid: 0
Pid: 43048
PPid: 3701
TracerPid: 0
Uid: 1000    1000    1000    1000
Gid: 1000    1000    1000    1000
FDSize: 256
Groups: 4 20 24 25 27 29 30 44 46 100 107 1000
NSTgid: 43048
NSpid: 43048
NSpgid: 43048
NSsid: 3392
Kthreadd: 0
VmPeak: 2692 kB
VmSize: 2692 kB
VmLck: 0 kB
VmPin: 0 kB
VmHWM: 1612 kB
VmRSS: 1612 kB
RssAnon: 100 kB
RssFile: 1512 kB
RssShmem: 0 kB
VmData: 224 kB
VmStk: 132 kB
VmExe: 4 kB
VmLib: 1752 kB
VmPTE: 44 kB
VmSwap: 0 kB
HugetlbPages: 0 kB
```


7. pmap

The pmap command displays the memory map of a process identified by its process ID (PID). It helps analyze how much memory is mapped to different regions.

Conclusion

Linux divides a process's virtual memory into different regions such as **code, data, heap, and stack**. By examining `/proc/<pid>/maps` and using pmap, we can understand the memory organization and address space of a running process.